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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|------------------------------------------------------------|------------------------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------------|--------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------|------------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| POLAR POCKET WPDS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                          |                                                            |                                          |                 |                                                                                                                                                                                                                          |            |                         |                                |                       | Pocket-FC-06-M                                                                                                   |                              | 0                                                                                        | March 18, 2026                                                                                                                                                  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                          |                                                            |                                          |                 |                                                                                                                                                                                                                          |            |                         |                                |                       | WPDS No.                                                                                                         |                              | Rev.                                                                                     | Date                                                                                                                                                            |  |
|  <b>Seaspan Vancouver Shipyards Co. Ltd.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                          |                                                            |                                          |                 |                                                                                                                                                                                                                          |            |                         |                                |                       | Applicable Standard(s)                                                                                           |                              | Lloyds Register - Rules for the Manufacture, Testing and Certification of Materials 2022 |                                                                                                                                                                 |  |
| Process/Mode                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                          |                                                            | Electrode (Wire) Classification          |                 |                                                                                                                                                                                                                          |            | Brand Name(s)           |                                |                       | Manufacturer(s)                                                                                                  |                              |                                                                                          |                                                                                                                                                                 |  |
| 1 FCAW/Semi-Auto(Hand)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                          |                                                            | AWS A5.20 E71T-1C/9C-J                   |                 |                                                                                                                                                                                                                          |            | Dual Shield Prime 71 LT |                                |                       | ESAB                                                                                                             |                              |                                                                                          |                                                                                                                                                                 |  |
| Material Designation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                          | Base material 1                                            |                                          |                 | Base material 2                                                                                                                                                                                                          |            |                         | Min. Preheat / Interpass Temp. |                       | 50°C for root and hot pass, fill/cap to be as per VSY Preheat and Interpass Temperature Requirements for Welding |                              |                                                                                          |                                                                                                                                                                 |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                          | EH 36 and all lower grades excluding A,B, D and E (Note 6) |                                          |                 | EH 36 and all lower grades excluding A,B, D and E (Note 6)                                                                                                                                                               |            |                         |                                |                       |                                                                                                                  |                              |                                                                                          |                                                                                                                                                                 |  |
| Delivery Condition(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                          | All except QT                                              |                                          |                 | All except QT                                                                                                                                                                                                            |            |                         |                                |                       |                                                                                                                  |                              |                                                                                          |                                                                                                                                                                 |  |
| Nominal Pipe Size                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                          | 500mm and above                                            |                                          |                 | 500mm and above                                                                                                                                                                                                          |            |                         | PWHT                           |                       | N/A                                                                                                              |                              |                                                                                          |                                                                                                                                                                 |  |
| Thickness or Dia                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                          | 3 mm - 60 mm                                               |                                          |                 | 3 mm - 60 mm                                                                                                                                                                                                             |            |                         | Max. Interpass                 |                       | 180°C                                                                                                            |                              |                                                                                          |                                                                                                                                                                 |  |
| Welding Position                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                          | All positions excluding Vertical down                      |                                          |                 |                                                                                                                                                                                                                          |            |                         | Joint Design                   |                       | Major/deep weld repairs                                                                                          |                              |                                                                                          |                                                                                                                                                                 |  |
| <p>Root gap for ceramic= Target 4-7mm (max 1.5 x T or 11mm, whichever smaller,</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                          |                                                            |                                          |                 |                                                                                                                                                                                                                          |            |                         |                                |                       |                                |                              |                                                                                          |  <p>U-shape profile with bottom radius of not less than 6mm and 25° wall</p> |  |
| TYPICAL 2G BUTTERING/WELDING SEQUENCE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                          |                                                            |                                          |                 |                                                                                                                                                                                                                          |            |                         |                                |                       | TYPICAL 3G BUTTERING/WELDING SEQUENCE                                                                            |                              |                                                                                          | REPAIR EXCAVATION PROFILE                                                                                                                                       |  |
| COMPLETE JOINT PENETRATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                          |                                                            | Welding Layer                            |                 | JOINT TYPE                                                                                                                                                                                                               |            | Back Purge              |                                | N/A                   |                                                                                                                  | Contact Tip to Work Distance |                                                                                          | 9.5-20 mm                                                                                                                                                       |  |
| <input checked="" type="checkbox"/> Back-gouged to sound metal<br><input checked="" type="checkbox"/> Welded onto backing<br><input type="checkbox"/> Welded from one side without backing<br><input type="checkbox"/> Welded both sides w/o back-gouging                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                          |                                                            | multi-layer                              |                 | <input checked="" type="checkbox"/> BUTT<br><input checked="" type="checkbox"/> CORNER<br><input checked="" type="checkbox"/> LAP<br><input checked="" type="checkbox"/> TEE<br><input checked="" type="checkbox"/> EDGE |            | Backing type            |                                | Ceramic/Steel         |                                                                                                                  | Interpass Cleaning           |                                                                                          | Grinding and Wire Wheel                                                                                                                                         |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                          |                                                            | One/Two side                             |                 |                                                                                                                                                                                                                          |            | Welding Technique       |                                | Stringer/Slight Weave |                                                                                                                  | Shielding Gas                |                                                                                          | 75% Argon / 25% CO2                                                                                                                                             |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                          |                                                            | Gun travel angle                         |                 |                                                                                                                                                                                                                          |            | Max. Bead Width         |                                | 16mm                  |                                                                                                                  | Gas Flow                     |                                                                                          | 16-25 LPM                                                                                                                                                       |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                          |                                                            | Pull w/ slight push on Vu                |                 |                                                                                                                                                                                                                          |            | Tungsten Electrode      |                                | N/A Ø:                |                                                                                                                  |                              |                                                                                          | 34-53 CFH                                                                                                                                                       |  |
| Method of steel preparation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                          |                                                            | Oxy fuel/Plasma cut   Grinding   Milling |                 |                                                                                                                                                                                                                          |            | No. of electrodes       |                                | 1                     |                                                                                                                  |                              |                                                                                          |                                                                                                                                                                 |  |
| BM Thickness Range mm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Layers / Passes          | Position                                                   | Electrode Size range (mm)                | Welding Process | Current type /Polarity                                                                                                                                                                                                   | Consumable | Current (A)             | Voltage (V)                    | WFS (IPM)             | Travel Speed (mm/min)                                                                                            | Heat Input kJ/mm             |                                                                                          |                                                                                                                                                                 |  |
| 3 ≤ T < 15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Root (Ceramic)           | All ex. Vd                                                 | 1.2 & 1.4                                | FCAW            | CV/DC+                                                                                                                                                                                                                   | E71T-1M/9M | 130 - 250               | 19 - 25                        | 170 - 300             | 55 - 130                                                                                                         | 1.5 - 2.6                    |                                                                                          |                                                                                                                                                                 |  |
| 15 ≤ T ≤ 30                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                          |                                                            |                                          |                 |                                                                                                                                                                                                                          |            |                         |                                |                       |                                                                                                                  | 1.5 - 2.6                    |                                                                                          |                                                                                                                                                                 |  |
| 3 ≤ T < 15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Buttering/ Hot/Fill/Ca p | All ex. Vd                                                 | 1.2 - 1.6                                | FCAW            | CV/DC+                                                                                                                                                                                                                   | E71T-1M/9M | 110 - 485               | 16 - 38                        | 124 - 500             | 57 - 640                                                                                                         | 0.5 - 1.8                    |                                                                                          |                                                                                                                                                                 |  |
| 15 ≤ T ≤ 30                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                          |                                                            |                                          |                 |                                                                                                                                                                                                                          |            |                         |                                |                       |                                                                                                                  | 0.5 - 2.4                    |                                                                                          |                                                                                                                                                                 |  |
| Note 1: Heat Input (kJ/mm) = [V x A x 60] / [Travel Speed (mm/min) x 1000]<br>Note 2: Grind groove joint and adjacent surfaces to bright metal prior to welding to remove all traces of paint, primer, scale, rust, moisture and any other contaminants. Wire brush, grinding to be used for interpass cleaning.<br>Note 3: Travel angle = 5-10 Pull / slight push should be used for Vertical up welding position.<br>Note 4: Amps and Volts are to be set by the amp/volt meter at the workpiece.<br>Note 5: This WPS qualifies both buttering and repairing welds where the gap is less than or equal to 16 mm or 1.5 x thickness, whichever is smaller.<br>Note 6: As required, gouging may be performed completely through the thickness. Excavated surfaces are to be re-beveled to the original joint configuration. Weld buttering may be performed on one or both faces of the bevel as required.<br>Note 7: Finished/ground buttered surface shall be successfully examined with MPI or LPI (whichever applicable) prior to application of the final groove weld. Make sure all traces of MPI/LPI consumables are cleaned from the weld joint prior to welding of the main joint.<br>Note 8: Welding of the normal strength hull structure steel to normal strength hull structure steel (Grade A,B,D and E) using Dual Shield Prime 71 LT is subject to special agreement with Lloyds Register.<br>Note 9: For the root pass on the ceramic backing, it is recommended to use 1.2mm (0.045") wire size with max. 180 amps. |                          |                                                            |                                          |                 |                                                                                                                                                                                                                          |            |                         |                                |                       |                                                                                                                  |                              |                                                                                          |                                                                                                                                                                 |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                          |                                                            |                                          |                 |                                                                                                                                                                                                                          |            |                         |                                |                       | Engineer Stamp                                                                                                   |                              |                                                                                          |                                                                                                                                                                 |  |
| Reference WPS No. FC-CS-R-01 (Rev. 2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                          |                                                            |                                          |                 |                                                                                                                                                                                                                          |            |                         |                                |                       |                                                                                                                  |                              |                                                                                          |                                                                                                                                                                 |  |